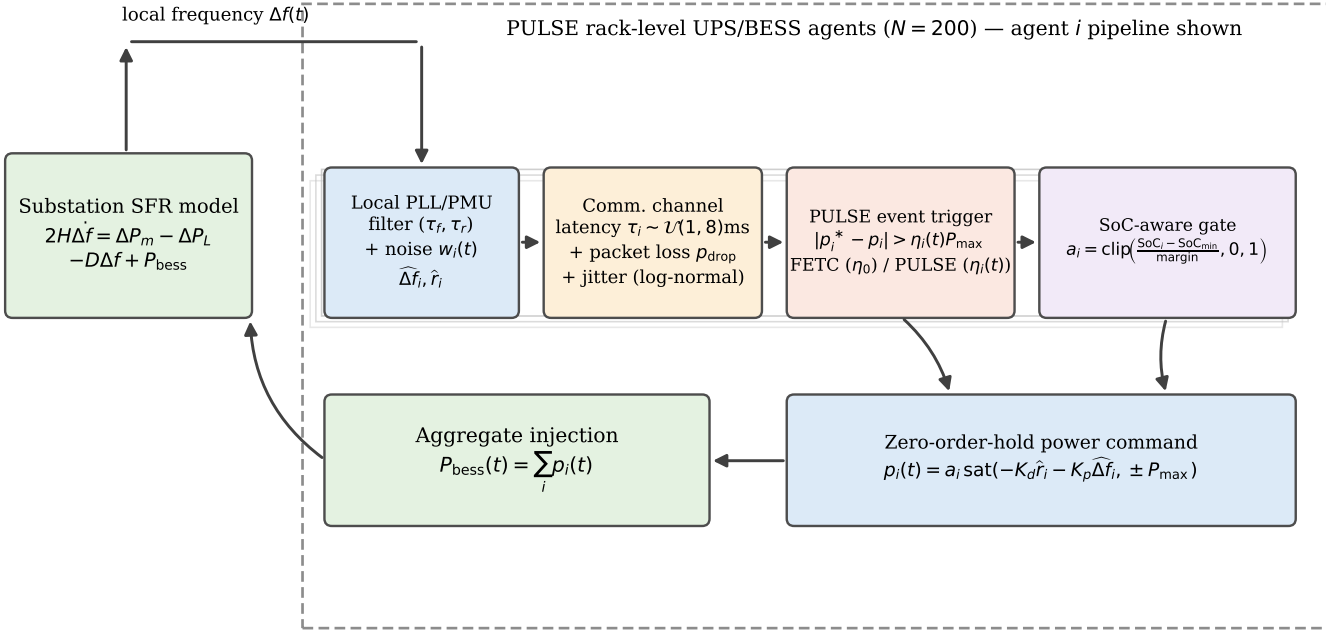
local frequency $\Delta f(t)$
PULSE rack-level UPS/BESS agents ($N = 200$) — agent i pipeline shown
Substation SFR model
$2H\dot{\Delta f} = \Delta P_m - \Delta P_L$
$-D\Delta f + P_{\text{bess}}$
Local PLL/PMU
filter (τ_f, τ_r)
+ noise $w_i(t)$
$\widehat{\Delta f}_i, \hat{r}_i$
Comm. channel
latency $\tau_i \sim \mathcal{U}(1, 8)$ms
+ packet loss p_{drop}
+ jitter (log-normal)
PULSE event trigger
$|p_i^* - p_i| > \eta_i(t) P_{\max}$
FETC (η_0) / PULSE ($\eta_i(t)$)
SoC-aware gate
$a_i = \text{clip}\left(\frac{\text{SoC}_i - \text{SoC}_{\min}}{\text{margin}}, 0, 1\right)$
Zero-order-hold power command
$p_i(t) = a_i \, \text{sat}(-K_d \hat{r}_i - K_p \widehat{\Delta f}_i, \pm P_{\max})$
Aggregate injection
$P_{\text{bess}}(t) = \sum_i p_i(t)$